

Hashing Concepts & Hash Functions — Data Structures

Definition: Hashing is a technique that transforms a search key k into a table index $h(k)$ using a **Hash Function**, enabling average constant time $O(1)$ insertion, deletion, and lookup operations in a **Hash Table**.

1. Detailed Technical Explanation

Search Key (e.g., 108) $\rightarrow$ [Hash Function $h(k) = k \% 10$] $\rightarrow$ Index 8 in Hash Table

1. The Load Factor (α)

The load factor represents the density of items stored in a hash table of size m with n keys:

Load Factor (α) = n / m (Number of Elements / Table Size)

- In Open Addressing, $\alpha \leq 1$. In Separate Chaining, α can exceed 1.

2. Common Hash Functions

1. Division Method (Modulo Arithmetic)

$$h(k) = k \bmod m$$

- **Rule:** m should be a **Prime Number** not close to a power of 2 or 10 to minimize clustering.
- *Example:* For table size $m = 11$ and key $k = 47$:
 $h(47) = 47 \bmod 11 = 3$

2. Mid-Square Method

1. Square the key: k^2 .
2. Extract the middle r digits of the square as the index (where table size $m = 10^r$).
 - *Example:* For key $k = 31$ and $m = 100$:
 $k^2 = 31^2 = 0961 \rightarrow$ Middle digits = 96 $\rightarrow$ Index = 96

3. Folding Method

Divide the key digits into equal parts of size r , and sum them together (ignoring overflow carry).

- **Fold-Shift:** Directly sum parts (e.g., Key 123456 into $12 + 34 + 56 = 102 \rightarrow 02$).
- **Fold-Boundary:** Reverse boundary parts before summing.

4. Multiplication Method

$h(k) = \text{floor}(m * (k * A \bmod 1))$ where $0 < A < 1$ (Knuth recommends $A \approx 0.6180339887$)

- *Advantage:* The choice of table size m is not critical (can be a power of 2).

3. Properties of a Good Hash Function

1. **Uniform Distribution:** Distributes keys evenly across all table slots to minimize collisions.
2. **Deterministic:** Must always compute the same index for the same input key.
3. **Fast Computation:** Computable in **$O(1)$** time.

4. Quick Recall Flow

Key $\rightarrow$ Hash Function $h(k) \rightarrow$ Index $[0..m-1]$ | Functions: Division ($k \% m$), Mid-Square (middle of k^2), Folding (split & sum)